Question 1

1. a) i)

$$f = \left(\overline{A + \overline{B}}\right) + \left(\overline{\overline{A} + B}\right)$$

$$= \overline{A}B + A\overline{B} = A \oplus B$$

[4]

ii)

$$g = \overline{\overline{\overline{AB}C}(B \oplus C)} = \overline{AB}C + \overline{B \oplus C}$$

$$= (\bar{A} + \bar{B})C + BC + \bar{B}\bar{C} = (\bar{A} + \bar{B} + B)C + \bar{B}\bar{C}$$

$$= C + \bar{B}\bar{C} = C + \bar{B}$$

[4]

b) (i)

AB \ CD	00	01	11	10
00	1	0	1	1
01	1	0	0	1
11	1	0	0	1
10	1	1	X	1

$$y = \overline{D} + A\bar{B} + \bar{B}C$$

[4]

(ii)

AB \ CD	00	01	11	10
00	1	0	1	1
01	1	0	0	1
11	1	0	0	1
10	1	1	X	1

$$y = \left(A + C + \overline{D}\right)\left(\overline{B} + \overline{D}\right)$$

[4]

Here, 2 marks for each of the Karnaugh maps with correct grouping, and 2 for each of the final expressions.

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c)

$$
\begin{aligned}
& AB + \overline{A}C + BC \\
= \; & AB + BC(A + \overline{A}) + \overline{A}C \\
= \; & AB + ABC + \overline{A}BC + \overline{A}C \\
= \; & AB(C + 1) + \overline{A}C(B + 1) \\
= \; & AB + \overline{A}C
\end{aligned}
$$

[4]

d)

Decimal	Hexadecimal	Binary	BCD
7244	**1C4C**		
		0000 0000 1101 0111	**(0010 0001 0101)** $_{\mathbf{BCD}}$ (As decimal is $(215)_{10}$)
14334	37FE		
-2417	**F68F**		

Give 2 marks per answer.

[8]

e)

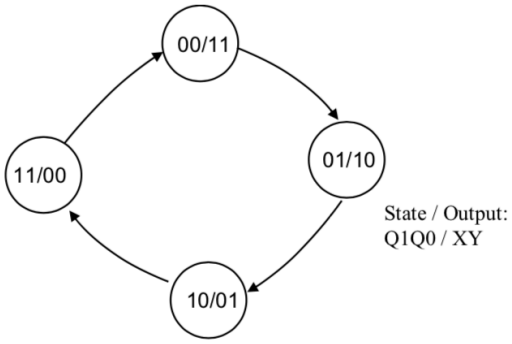
C	B	A	P	Q
0	0	0	0	0
0	0	1	1	1
0	1	0	1	0
0	1	1	1	1
1	0	0	0	0
1	0	1	1	1
1	1	0	1	1
1	1	1	1	1

Give 2 marks for correct use of inputs A, B, and C in table, and 2 marks each for P, Q outputs.

[6]

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f) (i) Moore state diagram:



Give 1 mark for correct number of states and connections, and 2 for state and output labels.

[3]

(ii) State transition table:

Current state	Next state	Output	
		X	Y
00	01	1	1
01	10	1	0
10	11	0	1
11	00	0	0

Give 1 mark for basic table, 1 for the correct input/output state transitions, 1 for the correct outputs

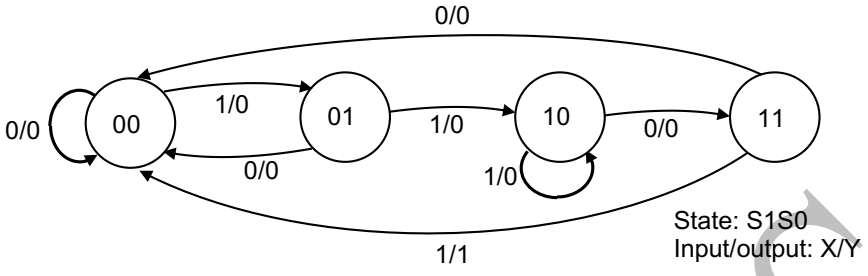
[3]

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Question 2

a) (i)

Mealy diagram for the FSM:



[8]

Give 2 marks for correct number of states, 3 for correct labelling, 3 for interconnections.

(ii)

State transition table for the FSM:

Current state		Input	Next state		Output
S1	S0	X	S1+	S0+	Y
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	0	0
0	1	1	1	0	0
1	0	0	1	1	0
1	0	1	1	0	0
1	1	0	0	0	0
1	1	1	0	0	1

[6]

Give 2 marks each for $S1^+$, $S0^+$ and Y.

iii)

The transition table for the D flip-flop is the following:

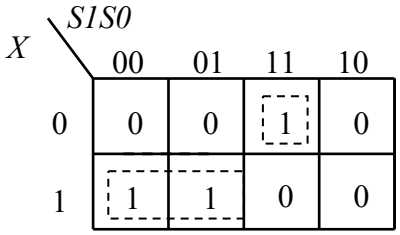
Transition	*D*
$0 \rightarrow 0$	0
$0 \rightarrow 1$	1
$1 \rightarrow 0$	0
$1 \rightarrow 1$	1

Give 1 mark for seeing this.

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For S1+:

K-map for $S1^+$:

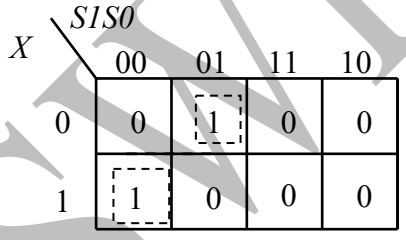


$$S1^+ = \overline{S1}X + S0S1\bar{X}$$

Give 2 marks for k-map and 1 for expression.

For S0+:

K-map for $S0^+$:



$$S1^+ = \overline{S0S1}X + S0\overline{S1}\bar{X}$$

Give 2 marks for k-map and 1 for expression.

For Y:

Y can be found directly from the state transition table:

$$Y = S1.S0.X$$

Give 1 mark for this.

[8]

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b) Boolean expressions for H, E and L are shown below:

$$H = A_3\overline{B}_3 + \left(\overline{A_3 \oplus B_3}\right)A_2\overline{B}_2 + \left(\overline{A_3 \oplus B_3}\right)\left(\overline{A_2 \oplus B_2}\right)A_1\overline{B}_1 + \left(\overline{A_3 \oplus B_3}\right)\left(\overline{A_2 \oplus B_2}\right)\left(\overline{A_1 \oplus B_1}\right)A_0\overline{B}_0$$

$$E = \left(\overline{A_3 \oplus B_3}\right)\left(\overline{A_2 \oplus B_2}\right)\left(\overline{A_1 \oplus B_1}\right)\left(\overline{A_0 \oplus B_0}\right)$$

$$L = \overline{A}_3B_3 + \left(\overline{A_3 \oplus B_3}\right)\overline{A}_2B_2 + \left(\overline{A_3 \oplus B_3}\right)\left(\overline{A_2 \oplus B_2}\right)\overline{A}_1B_1 + \left(\overline{A_3 \oplus B_3}\right)\left(\overline{A_2 \oplus B_2}\right)\left(\overline{A_1 \oplus B_1}\right)\overline{A}_0B_0$$

Alternatively, given H and E as above, we have: $L = \overline{H}.\overline{E}$

Give 3 marks for H and L, and 2 for E

[8]

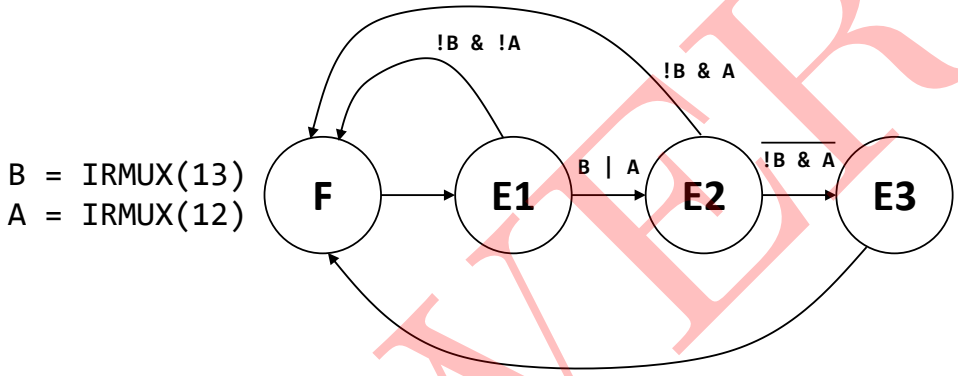
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3. Figures 3.1 and 3.2 show a non-pipelined CPU and its RAM. The CPU state machine has 4 distinct states: F, E1, E2 and E3. The CPU instruction set is defined in Figure 3.3. You may assume that all CPU registers are initially 0.

From its start the CPU executes, in sequence, `LDI`, `LDA` and `LDN` instructions as determined by the program in Figure 3.4. Figure 3.5 shows how the CPU state changes during the first 10 cycles of this program executing.

a) Draw the state transition diagram of the CPU state machine. Indicate conditional state transitions as boolean expressions of `IRMUX` signals, assuming no instructions other than as in Figure 3.3. [6]

Calculated example.The state machine is similar to a smaller one used before in lab work, but E3 is new, as is the calculation:
1 mark for each transition.



b) State the number of cycles delay between `Addr` changing, and the corresponding change in each of `Dout`, `IR`, `IRMUX`. Hence explain why the 4 DECODE signals from `IRMUX` could not come from `IR` or `Dout`. [6]

New analysis related to an example seen before:

1 cycle delay from `Addr` to `Dout` and `IRMUX` (2 marks).
2 cycles delay from `Addr` to `IR` (1 mark).

`IRMUX` is needed to hold the instruction word for use in the DECODE block throughout execution from E1 - E3.
`IR` is not valid during E1.
`Dout` is not valid during E2,E3. (3 marks).

c) For each of the states E1, E2, E3 of the `LDN` (0x2100) instruction state the contents of Dout, and describe how these relate to the `LDN` operation. [6]

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Analysis of new example. Similar hardware with LDI, LDA instructions has been used in lab work:
1 mark for each value, 1 mark for each description.
E1: 0x2100. LDN instruction word.
E2: 0x0101. Contents of 0x100.
E3: 0x0246. Contents of 0x101 to be loaded into Acc.

d) Name the memory addressing modes implemented by `LDA` and `LDN` instructions respectively, and state two advantages of immediate offset addressing, as used in the ARM CPU, over `LDA` and `LDN`. [4]

Bookwork:
`LDA` implements direct addressing (1 mark).
`LDN` implements indirect addressing (1 mark).
(1) Immediate offset addressing in the ARM CPU performs indirection without needing an additional memory read (1 mark).
(2) It allows offset addressing to be used in an arbitrary sized memory space (1 mark).

e) Figure 3.6 shows a 16 bit binary adder Σ with carry in and out `cin` and `cout` implementing single cycle two operand 16 bit arithmetic instructions. Two sequential cycles 0, 1 implement 32 bit subtraction: `R0:R1 := R0:R1 - R2:R3`.

i) State, with reasons, the values of `N` and `cin` in cycles 0 and 1. [6]

ii) State, using `X` for don't care, the values of `A` and `B` in each cycle. [2]

Application of bookwork:
cycle 0 cin=1 to implement 2's complement negation (2 marks)
cycle 1 cin = CARRY to implement subtraction carry (2 mark)
Both cycles N = 1 to invert one operand for 2's complement negation (2 marks)
Analysis of new example:

Cycle	A (1 mark)	B (1 mark)
0	X	1
1	1	0

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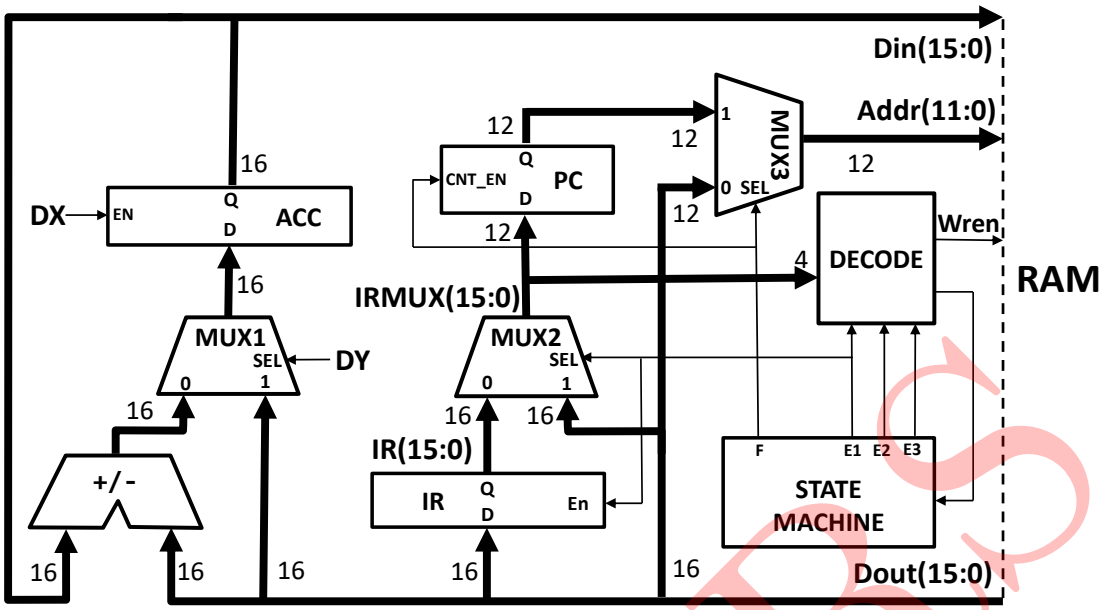


Figure 3.1: CPU Block Diagram

$\texttt{Wren}_N$	$\texttt{Addr}_N$	$\texttt{Din}_N$	$\texttt{Dout}_{N+1}$	Operation
0	a	X	$\texttt{Mem}_{16}[a]$	Read
1	a	d	X	$\texttt{Mem}_{16}[a]$:= d

Figure 3.2: RAM operation: S_N indicates signal S during cycle N

Mnemonic	IW(15:12)	Operation
LDI	0b0000	Acc := IW(11:0)
LDA	0b0001	Acc := $\texttt{Mem}_{16}$[IW(11:0)]
LDN	0b0010	Acc := $\texttt{Mem}_{16}$[$\texttt{Mem}_{16}$[IW(11:0)]]

Figure 3.3: Instructions: IW(15:0) is the instruction word

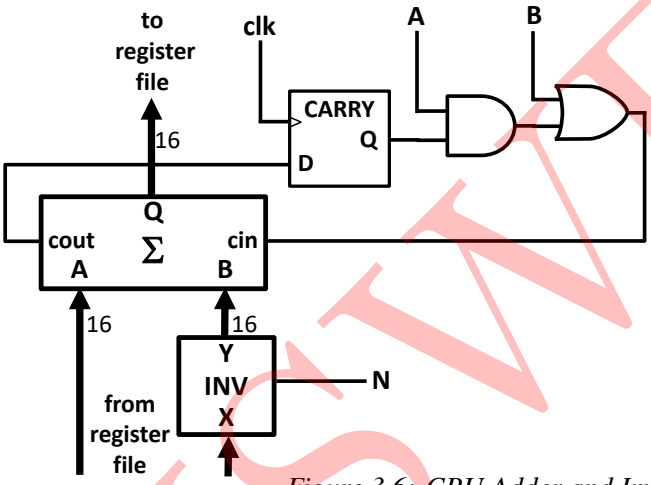
Addr	$\texttt{Mem}_{16}$[Addr]	Assembler
0x000	0x0123	LDI 0x123
0x001	0x1100	LDA 0x100
0x002	0x2100	LDN 0x100
0x100	0x0101	
0x101	0x0246	

Figure 3.4: Initial RAM contents for program: all other locations are 0

Cycle	0	1	2	3	4	5	6	7	8	9
PC	0	1	1	2	2	2	3	3	3	3
State	F	E1	F	E1	E2	F	E1	E2	E3	F

Figure 3.5: States during first 10 cycles executing the program in Figure 3.4

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INV block		
N	X(i)	Y(i)
0	x_i	x_i
1	x_i	$\overline{x_i}$

Figure 3.6: CPU Adder and Invert block

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